

The advisory a farmer never gets.

AgriMitra Climate reads a specific field — its weather, its satellite vegetation, its soil, its crop — and returns one decision for the week ahead, with every assumption shown.

86%

of Indian farmers hold under 2 hectares

1:1000

extension officers to farmers

15–25%

of yield lost to pests and disease

₹1.4L cr

upper bound of that loss, annually

THE USER

Ravi farms 0.8 hectares in Alwar.

He knows rain is coming. What he does not know is whether, given his loamy soil, his limited irrigation, the state of his wheat this week and the forecast, he should irrigate now or wait.

- The nearest extension officer covers roughly a thousand farms like his.
- Weather apps give a forecast, not a decision.
- A disease scanner names a disease, then stops.
- Advice that arrives on WhatsApp is generic to the district, not to the field.

Rain is coming. So — do I irrigate today, or lose the window?

Farmers who receive early, accurate guidance cut crop losses by **30–60%**. The signal exists. It does not reach the field in a form anyone can act on.

0.38 ha

Average marginal holding nationally — decisions are made at this scale.

68.5%

Of farm households are marginal, under one hectare.

5 states

UP, Bihar, MP, Maharashtra and Andhra hold more than half of them.

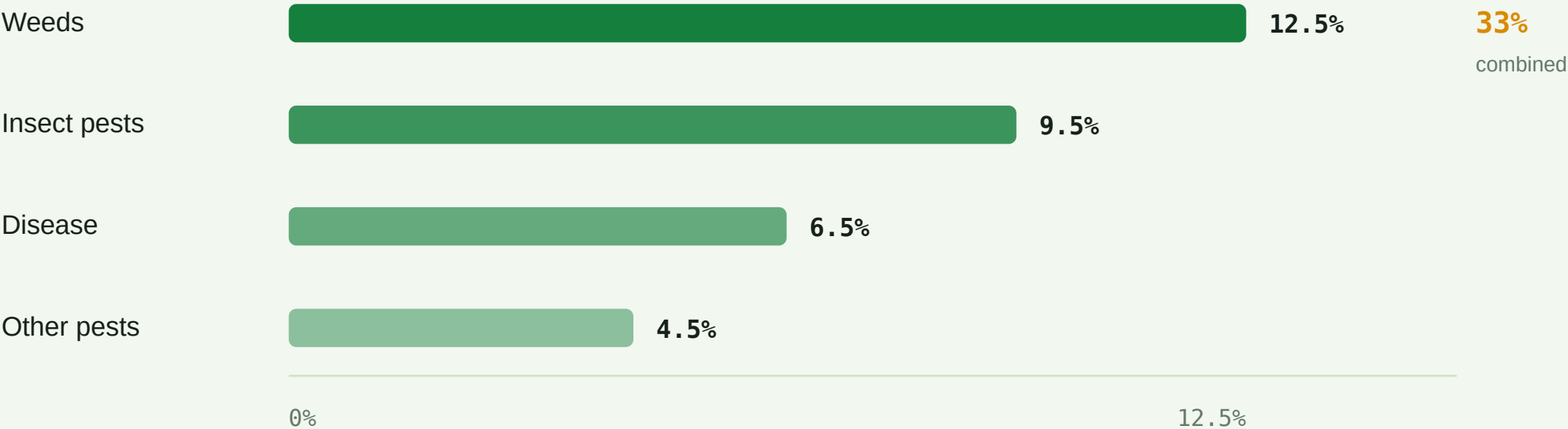
24%

Of cultivable land, worked by those 68.5% of households.

PROBLEM • MAGNITUDE

A third of the crop is lost before it leaves the field.

Share of Indian production lost, by cause. Every one of these is visible in a leaf or a soil-moisture reading days before it becomes a loss.



Shares are of total production loss attributed to pests in Indian agriculture.

Six apps, no answer.

The data a farmer needs already exists. It arrives in fragments, none of which knows about the others, and none of which knows the field.

Weather is not advice

A 7-day forecast is an input. Turning it into “irrigate Thursday, not today” needs the soil, the crop stage and the water available.

Diagnosis stops at the name

A scanner returns “Early blight” and ends. It never reaches the risk score, and never changes what the farmer is told to do next.

Satellite stays with agronomists

NDVI is routine in enterprise platforms and absent from the farmer's phone.

Advice is district-wide

Two farms in one district with different soil and irrigation get identical guidance.

Confidence is never shown

Tools present estimates as measurements. A farmer cannot tell which number to trust.

States cannot share models

Every state rebuilds its own crop dataset and risk logic. Nothing is interoperable.

The pattern

Each product owns one layer well and hands the farmer the integration problem.

The cost

Six apps, six logins — and still no answer for this field, this week.

The opening

Nobody is competing on the join between the layers.

THE SOLUTION

One field. Five signals. One decision.

Deterministic logic computes the score. Gemini explains it. The model never invents a measurement — an architectural rule, enforced in code.

- 01 **Field profile** — location, crop, season, soil, irrigation. Entered once; GPS fills district and state.
- 02 **Live weather** — current conditions and a 7-day forecast for that coordinate.
- 03 **Satellite vegetation** — Sentinel-2 NDVI reduced over the field via Google Earth Engine.
- 04 **Leaf scan** — a photograph routed to MobileNetV2, escalating to Gemini multimodal below the confidence threshold.
- 05 **Risk score** — five weighted factors, and a written note for every factor that had no evidence.
- 06 **Advisory** — Gemini narrates the computed numbers in English, हिन्दी, ગુજરાતી or తెలుగు.

RISK ENGINE

Water 0.30 · Heat 0.20 · Disease 0.20 · Rainfall 0.15 · Vegetation 0.15

When the weather service is unreachable, heat and rainfall are **dropped from the weighted score** and the remaining weights renormalised — never silently read as zero.

SHOWN TO THE FARMER

“No leaf analysis on record; disease risk uses the neutral baseline (30/100). Scan a leaf to replace it.”

Deterministic first

The score is arithmetic over five named weights — reproducible and auditable.

Gemini second

The model narrates the computed numbers. It cannot change a score.

Farmer last

Plain language, four languages, every assumption written beneath the number.

WHAT MAKES IT DIFFERENT

It tells the farmer what it does not know.

Agricultural advice moves money and food. A confident wrong answer is worse than an admitted gap — so provenance is a first-class field in the API, not a footnote in the UI.

— Every measurement carries

`is_live` ,

`degraded` and

`is_demo` — the types make them non-optional, so a screen cannot render a number without the means to label it.

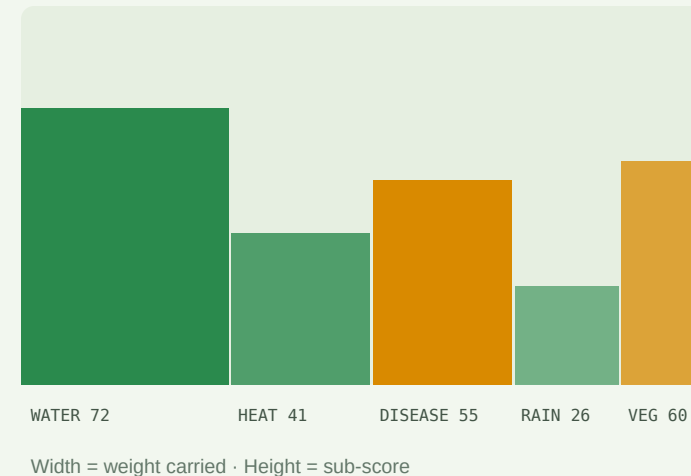
— Demo satellite data says

“Demo dataset — not a live satellite observation.”

— With no Gemini key, the advisory is labelled rule-based rather than passed off as model output.

— Mandi prices return a 503 when unconfigured. Never an estimate.

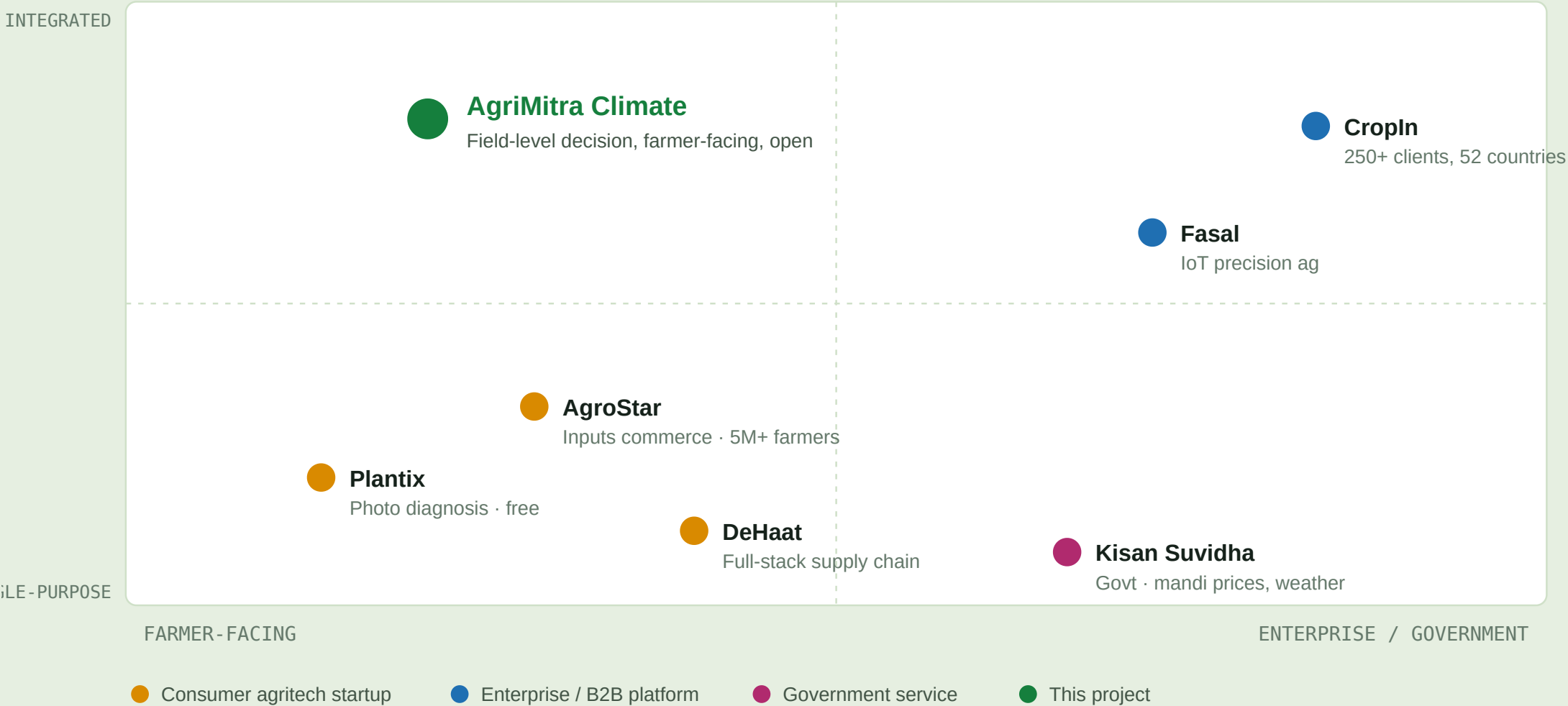
— The leaf scanner names the 14 crops it was trained on, and warns before you photograph a crop outside them.



The signature view. It answers “what is driving this number” — which five independent progress bars cannot. A factor with no evidence is absent, not drawn at zero.

Everyone owns one layer. Nobody owns the decision.

Horizontal: who the product serves. Vertical: whether it answers a single question or produces a field-level decision.



Placement reflects each product's primary positioning, not its full feature list.

The landscape, honestly.

These are strong products. The gap is not that they are bad — it is that none of them closes the loop from satellite and soil to a decision the farmer can act on, and none is built to be shared between states.

PRODUCT	TYPE	CORE STRENGTH	SCALE	GAP THIS LEAVES
Plantix	STARTUP	Best-in-class photo disease diagnosis, free, many Indian languages	—	Diagnosis only; no soil, satellite or field risk
AgroStar	STARTUP	Content-led input commerce; farmers post crop photos for advice	5M+ farmers	Advisory serves product sales; not a neutral decision layer
DeHaat	STARTUP	Full-stack inputs, advisory and market linkage across 12 states	\$700M+ val.	Supply-chain first; satellite risk is not the product
CropIn	ENTERPRISE	Satellite + ML farm intelligence; Gemini partnership since 2024	52 countries	Sold to agribusiness, not reachable by a 0.8 ha farmer
Fasal	ENTERPRISE	IoT sensors driving hyper-local irrigation and spray timing	—	Requires hardware per farm; capital cost excludes smallholders
Kisan Suvidha	GOVERNMENT	Official mandi prices, weather and advisories, free	National	Reference data, not analysis; nothing is field-specific
AgriMitra Climate	THIS PROJECT	Satellite + soil + weather + leaf → one explainable field risk score	Hackathon build	Unproven at scale — see roadmap

Indian agritech raised **\$241M across 60 rounds in 2025**; the market is projected to grow from **\$974M (2025) to \$2.52B by 2034**.

Google AI, load-bearing.

Each service below has a caller and a user-reachable code path. Where credentials are absent the feature degrades to a labelled fallback — it is never faked to pad a list.

REASONING

Gemini API · Vertex AI
Advisory, assistant, district interventions, crop-ranking explanations. Structured output, temperature 0.2, and a system instruction forbidding invented measurements. Vertex gives a second auth path using service-account IAM, so Cloud Run holds no API key.

VISION

Gemini multimodal
Second opinion whenever the local classifier falls below 0.70 — which is every crop outside its 14 trained ones, i.e. most Indian staples.

GEOSPATIAL

Earth Engine · Maps
Sentinel-2 NDVI reduced over a 150 m buffer, cloud-filtered to 30%. Maps Platform renders the district risk layer.

LANGUAGE

Speech & Translation
Cloud Speech-to-Text and Text-to-Speech for the browsers with no Web Speech API — Android WebView, most in-app browsers. Translation localises the scheme directory.

DATA

BigQuery · Cloud Storage
Every computed risk is appended to a warehouse time series, including which factors carried evidence. Leaf photos are retained as the evidence behind a diagnosis.

PLATFORM

Cloud Run · Secret Manager
Three services, non-root images, health checks. Secrets injected at runtime. Pub/Sub fans out NDVI refresh; Redis caches weather and satellite reads.

Public data: `data.gov.in` Agmarknet mandi prices · Open-Meteo forecast and historical archive · OpenStreetMap reverse geocoding · Sentinel-2 via Earth Engine.

Degrades honestly

No Gemini key means a labelled rule-based advisory, not a silent fallback.

No key on the box

Vertex AI uses the Cloud Run service account's IAM identity.

Costs bounded

Redis caches weather, NDVI and translations across neighbouring farms.

THE THESIS

A crop dataset is a public good. Ship it as one.

The challenge asks for an interoperable network that lets Indian states share agricultural data models. Two things in this system already *are* portable models — they simply need publishing as such.

—

The crop dataset

— ten crops with season, soil, water, temperature and rainfall envelopes.

—

The risk model

— five named weights, four bands, and the rules for dropping a factor with no evidence.

Published as versioned JSON Schema behind a federation API, Rajasthan's crop model becomes something Maharashtra can adopt, critique or replace — without a bilateral integration.

FEDERATION, PRIVACY FIRST

—

Aggregates only.

No farm record, no farmer record, no coordinate finer than a district centroid crosses a boundary.

—

A k-anonymity floor.

A district with fewer than ten farms is omitted — “1 farm, CRITICAL, Wheat” identifies a person.

—

Signed manifests.

A peer verifies before ingesting, so a hostile node cannot poison another state's dashboard.

—

Enforced by test

, not by policy prose.

Open licence · open standard · documented · no PII · platform independent — the five Digital Public Good criteria, each mapped to where the code satisfies it.

Working software, not slideware.

26

API endpoints, OpenAPI-
documented

90

automated tests across two
services

10

screens, 4 languages, light and
dark

100

seeded farms across 10
districts

Running end to end

- Farm registration with GPS → district resolution → risk in one flow, against MongoDB Atlas.
- Live Open-Meteo weather and measured seasonal rainfall from the historical archive.
- Leaf scan: 20 ms rejection of non-leaves, 160–260 ms inference, 612 ms for a 3600 px photo.
- District roll-up aggregating 100 farms into risk bands and dominant crops.

Bugs found and fixed by testing, not reading

- A confident *healthy* leaf produced a disease risk of 95/100 .
- The leaf detector rejected every diseased colour — rust, blight, necrosis — as “not a plant”.
- Crop rainfall was compared in mm/hour against a mm/season requirement, so that criterion never scored.
- A dead weather API was read as “0 °C, no rain” instead of as missing.

Verified against Atlas

Farmer → farm → risk end to end on a live cluster with 100 seeded farms.

Verified against live APIs

Open-Meteo, OpenStreetMap and Agmarknet all called for real.

Gated on every change

ruff, pytest, tsc, eslint, next build — all five green or the phase is not done.

ROADMAP

What comes next, in order.

Ranked by value to a farmer or an officer, divided by effort.

NEXT	WHAT IT UNLOCKS	EFFORT
Federation layer	Published schema plus peer endpoints — the interoperable network the challenge asks for	1 day
FAO crop coefficients	Replaces prototype constants with citable FAO-56 values	2 hours
Regenerative scoring	Rotation-aware ranking: a legume after a cereal scores above the cereal repeated	0.5 day
Vertex AI AutoML	Next-season risk forecast trained on the BigQuery history, shown beside the computed score, never replacing it	1 day
Indian crop disease model	Wheat, rice, mustard, chickpea — the crops the current checkpoint has never seen	Ongoing
ISRO Bhuvan · IMD	Indian sovereign satellite and meteorological sources behind the existing provider interface	1 day

TIMING

Why this is buildable now and was not three years ago.

Three curves crossed. Satellite analysis became free at the field scale, multimodal reasoning became an API call, and the farmer's device became capable of both.

THEN: ENTERPRISE ONLY

Satellite went free and granular

Sentinel-2 revisits every five days at 10 m, and Earth Engine reduces it over a single field on demand. What needed an agronomy contract in 2020 is one API call per farm.

THEN: BESPOKE MODELS

Multimodal reasoning became callable

A leaf photograph plus structured farm context in one prompt, answered in Hindi or Gujarati, is a 2024-onwards capability. Before that it meant a trained model per crop per language.

THEN: FEATURE PHONES

The device caught up

Rural smartphone penetration now carries a camera good enough for lesion detection and a browser capable of speech input — which is what makes a farmer-facing product possible at all.

\$974M → \$2.52B

Indian agritech market, 2025 to 2034, at 10.59% CAGR.

\$241M / 60 rounds

Raised across Indian agritech in 2025 alone.

86.1%

Of holdings are under two hectares — the segment every enterprise platform prices out.

THE PRODUCT

Ten screens. One thread running through them.

Every screen answers a question the previous one raised, and every number carries a label saying where it came from.

SCREEN	WHAT IT DOES	DETAIL
Register	Name, language, GPS. Reverse geocoding fills district and state so the farmer types neither.	4 languages
My Farm	The risk score as a soil core, plus a written note for every factor that had no evidence.	Home screen
Weather	Current conditions and seven days, with source and fetch time always visible.	Open-Meteo
Scan	A leaf photo. Names the 14 crops it can identify before you shoot, and warns if yours is not among them.	160–260 ms
Crops	Ranked suitability with a per-criterion breakdown; unknowns read “not assessed”, never as failures.	10 crops
Ask	Voice or text, grounded only in this farm's record. Returns “I cannot answer that” rather than guessing.	Web Speech
Districts	State roll-up, risk map and AI intervention priorities for an agriculture officer.	For the officer
Mandi & Schemes	Agmarknet prices for the crop actually grown, and the subsidy directory in the chosen language.	data.gov.in

Three services, one contract.

The web app never talks to a model or a satellite directly. Everything crosses one typed API, which is what makes the provenance guarantee enforceable rather than aspirational.

APPS/WEB

Next.js 15

10 routes, React 19, server components, 4 languages, light and dark.

SERVICES/API

FastAPI • domain

26 endpoints. Risk, suitability and district aggregation are pure functions with no I/O — which is why they are testable.

SERVICES/INFERENCE

FastAPI • vision

MobileNetV2 plus an HSV leaf detector; escalates to Gemini below 0.70 confidence.

MongoDB Atlas

System of record. A connection outage degrades stored-data routes to 503 and leaves the stateless ones serving.

BigQuery • Pub/Sub • Redis • GCS

Risk time series, async NDVI refresh, cached reads, and leaf photos kept as diagnostic evidence.

Gemini • Vertex AI

Advisory, chat, interventions, crop explanations and the vision fallback. Structured output only.

Earth Engine

Sentinel-2 NDVI per field, with a seeded demo provider behind the same interface, labelled `is_live: false`.

ADOPTION

The officer is the distribution channel.

A farmer-facing app with no distribution reaches nobody. The district view is not a second product — it is how the first one gets adopted.

1

One district

Seed one district's farms with an extension office. The officer already has the relationships an app does not.

Weeks 1–4

2

The officer's own view

District risk hotspots and intervention priorities give the officer a reason to keep the system current.

Weeks 2–8

3

Farmers pulled in

Farms the officer registers become farmer accounts. Adoption follows an existing trust relationship.

Weeks 4–12

4

The state adopts the model

Published crop and risk schema let a second district — then a second state — run its own node against the same standard.

Quarter 2+

Zero hardware

Nothing to install in a field — what separates this from IoT precision ag and its per-farm capital cost.

Zero sign-up

No account required to see a field. Authentication is deliberately out of scope on the farmer path.

Runs degraded

Every integration has a labelled fallback, so a district with no Gemini budget still gets the computed score.

What one district would be worth.

The arithmetic below is a model built from the sourced figures on slide 3, not a result. It is shown so the assumptions can be argued with.

2,340

farms in a district of Alwar's
size

15–25%

of yield currently lost to pests
and disease

30–60%

reduction where guidance
arrives early and accurately

4.5–15%

of yield recoverable, as the
product of those ranges

How to read this

- The 30–60% figure describes farmers who received early accurate guidance — it is not a claim about this product.
- Reaching it requires the advisory to be both timely and correct, which is what the roadmap's remaining items address.
- No revenue model is presented. A public good funded by a state department has different economics to a startup, and pretending otherwise would be dishonest.

What would have to be true

- A disease model trained on Indian staples — wheat, rice, mustard, chickpea — which the current checkpoint has never seen.
- Advisory quality validated against agronomists, not against a demo.
- Enough farms per district to clear the k-anonymity floor before any aggregate is published.
- An extension officer willing to keep one district's records current.

RISKS

What would break this, and what we did about it.

Every one of these is a real failure mode. Four already happened during the build and were fixed; the rest are open.

RISK	WHY IT MATTERS	RESPONSE	STATUS
Wrong advice at scale	Agricultural advice moves money. A confident error compounds across a district.	Deterministic scoring; Gemini cannot alter a number; every assumption written out.	Mitigated
A model outside its training set	The disease checkpoint covers 14 crops; nine of the ten we recommend are not among them.	The scanner names its coverage before you photograph and warns when your crop is outside it.	Mitigated
Silent data degradation	A dead weather API once read as "0 °C, no rain" and quietly changed the score.	Unavailable factors are dropped and the remaining weights renormalised, with a note shown.	Fixed
Re-identification via aggregates	A district with one farm publishes "1 farm, CRITICAL, Wheat" — which identifies a person.	k-anonymity floor of ten farms before any district appears in a federation response, enforced by test.	Designed
Connectivity in the field	The advisory is useless if it needs a live connection at the moment of decision.	Open — offline caching and a PWA shell are not yet built.	Open
Adoption without the officer	A farmer-facing app with no distribution reaches nobody.	Officer-first rollout, see slide 16. Unproven.	Open

Give the field a voice it can act on.

Not another weather app. Not another disease scanner. The first layer that reads a specific field across every signal available — and is honest about the ones it is missing.

For the farmer

A weekly decision, in their language, with the working shown.

For the officer

District risk hotspots and AI intervention priorities, from real aggregation.

For the state

A crop model and a risk model any other state can adopt, critique or replace.

Sources — Agriculture Census 2015–16 (86.1% of holdings under 2 ha; marginal holdings average 0.38 ha) · ICAR / CEEW on crop losses (15–25% of yield; ₹0.9–1.4 lakh crore annually; weeds 12.5%, insects 9.5%, disease 6.5%, other 4.5%) · extension coverage ≈ 1 officer per 1,000 farmers; early accurate guidance reduces losses 30–60% · Tracxn, Inc42 and AgroSpectrum India on funding and market size (\$241M across 60 rounds in 2025; \$974M in 2025 to \$2.52B by 2034) · company profiles from Tracxn, TechCrunch and Decentro. Figures describe the Indian market as reported by those sources; placements on slide 07 are our reading of each product's primary positioning.